



CASE STUDY

OISD/CS/2025-26/E&P/14

Dt.:30.10.2025

INTRODUCTION

Title: Fatal accident caused by tubing drop after the elevator was unlatched at the monkey board

Location: Onshore Workover Rig

Loss/ Outcome: Fatality of the Night Tool Pusher operating the driller console, Critical injury to a Rig-man

BRIEF OF INCIDENT:

A fatal accident occurred at Workover Rig. The incident took place during running-in operations when a go-gauge was stuck in a tubing-double (two joined 2-7/8" tubings). Initial efforts to release it by hammering with a 2" pipe were unsuccessful. The Night Tool Pusher and the crew lowered and placed the pin-end of the tubing double on the incline ramp to further attempt removal of the go-gauge from the tubing by hammering.

The topman pulled the tubing-double toward the racking fingers with the rope and unlatched it from the elevator. The tubing-double then slid down suddenly and uncontrollably. Its box end struck Night Tool Pusher near the driller console, causing a severe head injury. The tubing also hit a rig-man, who was descending the stairs, causing a serious injury by crushing him against the stair handrail, resulting in multiple rib fractures and broken left hand. The travelling block then descended uncontrollably, struck the box end of the tubing held on the slips, and its elevator and links hit the rig floor. Another rig-man engaged the brake. Fortunately, the travelling block's fall caused no further injury or damage.

OBSERVATIONS/ SHORTCOMINGS

1. During the incident, the information regarding the placement of the pin end on the incline ramp at the V-door was not conveyed to the topman. The topman was unaware of the activity taking place on the rig floor. He was engaged in conversation with another rig man at the monkey board. As a result, the necessary information about the placement of the pin end of tubing at the incline ramp and instructions not to open the elevator did not get communicated to him.
2. The topman used a rope to pull the tubing double which normally precedes the unlatching of elevator during tripping. No crew member on the rig floor noticed the topman's action of using the

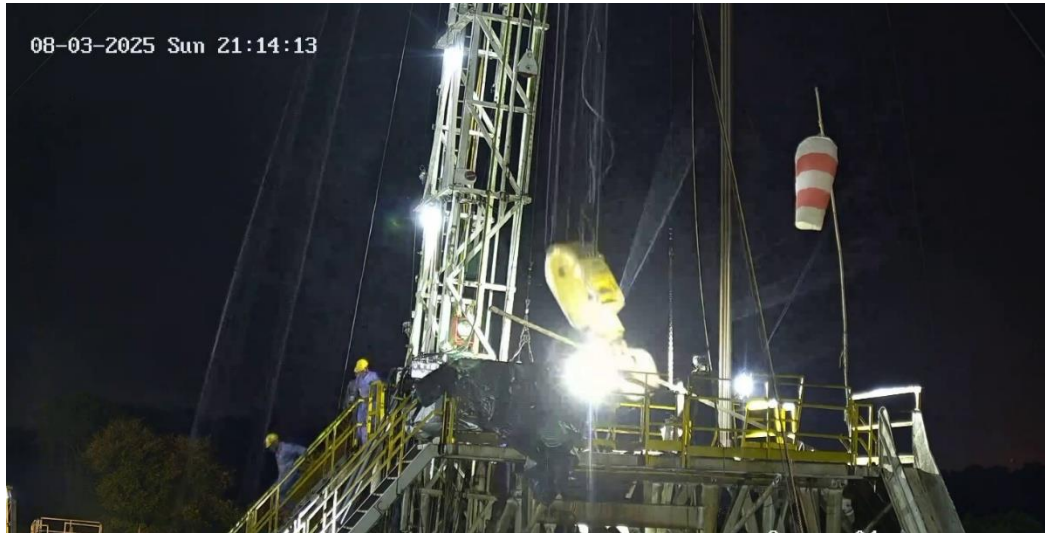
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rope to unlatch the elevator, and as a result, he was not alerted or stopped.

3. Observations noted from the CCTV recordings are:

- Rig crew members were using mobile phones during the workover operation on job.
- During the incident, the elevator was unlatched by the topman.

4. Twin stop device failed to function during the incident. The daily checklist indicated that the device was checked at the beginning of the shift. Rigor of the checking might not have been adequate.



Picture 1: Uncontrolled Fall of Travelling Block

5. The catwalk showed multiple impact marks and was misaligned with the substructure. Tubing-double, which had slid down the ramp, bounced over the V-door handrail and box end hit the tool pusher.

CONCLUSION / ROOT CAUSE

The root causes of the incident are as follows:

- A. The topman unlatched the tubing double from the elevator while the pin end was on the incline ramp due to communication gap.
- B. The travelling block descended uncontrollably and hit the rig floor as the twin stop device did not engage. This non-functionality of the twin stop device occurred presumably due to bypassing or not setting of the device.

RECOMMENDATIONS

1. **Communication protocol:**

- A clear communication protocol between the driller and the topman should be established.
- Effective means of two-way communication system between the monkey board and the rig floor shall be available for workover rigs.

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- Non-routine tasks like go-gauging should be carried out with utmost alertness among crew members, including the topman.

2. Unsafe Acts / Conditions:

- Use of mobile phones in the operational areas must be strictly prohibited due to the risk of fire and distraction of personnel.
- Organization should establish a periodic review of CCTV footage from rig operations to identify and address unsafe acts or conditions. Repeated offenders must be penalized.

3. **Twin stop Device:** As per regulation 42 (f) of Oil Mines Regulations-2017, the draw-works shall be provided with an automatic device which shall effectively prevent the travelling block from coming closer than two metres of the crown block on one end and crashing on the rotary table at the other end. The twin stop device should be physically verified before being marked 'checked' in the daily checklist.
4. **Catwalk orientation:** The catwalk should be correctly aligned with the substructure and fitted with a functional end stopper to prevent tubulars from jumping out. Damaged stoppers must be repaired or replaced immediately.

POSITION OF TUBING DOUBLE AND TRAVELLING BLOCK POST INCIDENT



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